Impact of Geopolitical Events on Global Crude Oil Prices Progress Report

Project Scope Update

The project scope remained unchanged and successfully addressed all challenges encountered. Its purpose is to examine how significant geopolitical events, such as armed conflicts, sanctions, and political instability affects global crude oil prices, and to create predictive models utilizing geopolitical and supply-side factors.

Data Sources

Three data sources have been successfully integrated into the project pipeline:

EIA Spot Prices (API): Monthly Brent crude oil spot prices are retrieved via the EIA open data API. The dataset covers January 2015 through December 2024 and returns date and price fields in JSON format, loaded directly into a pandas DataFrame.

EIA International Oil Production (API): Monthly crude oil production by country is retrieved via the EIA international API filtered by production, and country. Data covers seven major producers: Saudi Arabia, Russia, Iraq, UAE, United States, Iran, and Kuwait, spanning 2015 to 2024. Values are returned in thousand barrels per day and converted to million barrels per day.

ACLED Political Violence Events (CSV): Armed conflict and political violence events are loaded from locally stored ACLED CSV files covering 2015 to 2025. Records are filtered to political violence events only, parsed to monthly periods, and aggregated into event counts per country per month before merging with the production and price data. Full detailed table of all events is loaded to dig deeper into analysis.

Issues and Difficulties

Two significant data source changes were required after the initial proposal:

1. The originally planned geopolitical events source, GDELT, proved unsuitable for this project. After several days of downloading and processing over 100 million records, the data quality was unacceptably low, the GoldsteinScale severity metric was missing or zero in the majority of records, making conflict intensity analysis impossible. Data redundancy was high, with the same events reported multiple times across sources. A large proportion of rows were irrelevant to geopolitical analysis, referencing corporate entities, schools, and administrative bodies. The dataset required significant RAM to process and the coding effort to clean and filter it produced results with no analytical value. GDELT was replaced with ACLED, which

provides human-verified, researcher-curated conflict event data with reliable fields, free API and CSV access, and direct relevance to political violence in oil-producing regions.

2. The second change involved oil production data. The World Bank production dataset is published annually, which is insufficient for capturing the month-to-month supply volatility that follows geopolitical shocks. It was replaced with the EIA International API, which provides monthly country-level production data for all major producers from 2015 onward.